Work Integrated Learning Programmes Division
M.Tech (AIML/ DSE)
Machine Learning

Assignment - 1

Marks = 5

Objective

This Assignment is regarding the **Bike Sharing Demand Prediction Challenge**, where your goal is to predict the number of hourly bike rentals, using weather, time, and seasonal data. Through this problem, you will:

- Learn how to analyze a real-world dataset,
- Apply Linear Regression and its extensions,
- Understand nonlinearity and regularization,
- Evaluate models using a logarithmic error metric (RMSLE).

Dataset

- bike_train.csv – Training data with features and target
https://drive.google.com/file/d/18kpPgnoDfOfx3PRRZY3xbsLeWalNViYl/view?usp=drive_link
- bike_test.csv – Test data without target
https://drive.google.com/file/d/1HJlgPTkGlCzycSTnJNB0oJEfgjSV4o0f/view?usp=drive_link
- sampleSubmission.csv – Example format for predictions
https://drive.google.com/file/d/1g_hQQtI80Dz0NkGXG8ZioND9iVGTerd8/view?usp=drive_link

Target column: count (number of bikes rented per hour)

Evaluation Metric

You will be graded primarily on RMSLE:
RMSLE = $\text{sqrt}((1/n) * \Sigma (\log(\text{pred}+1) - \log(\text{actual}+1))^2)$

This function is not available in the sklearn library. You can use the below code to compute RMLSE:

```
def rmsle(y_true, y_pred):

   return np.sqrt(np.mean((np.log1p(y_pred) - np.log1p(y_true))**2))
```

Task Breakdown

Executing the Assignment on BITS Lab portal - 0.5 Marks

Exploratory Data Analysis (EDA) - 0.5 Marks

Q1. Examine dataset size, missing values, and feature types.

Q2. Visualize relationships between key features and the target variable (count).

Q3. Suggest which variables are likely to be most informative.

Feature Engineering - (Optional hints to improve performance)

Q4. You can try to derive features from datetime (hour, weekday, month, season), encode categorical variables, consider transformations to capture nonlinear trends to improve your model performance. If you do any of these, report it as answer to Q4. It is optional.

Regression Model - 1 Mark

Q5. Split data into training and validation sets and build a simple Linear Regression model.

Q6. To improve model performance, you may try to:

- Extend feature space using polynomial transformations (degree 2 or 3)
- Apply Ridge and Lasso regression on polynomial features, Tune the regularization strength (α).

Model Comparison and Interpretation - 1.5 Marks

Q7. Summarize all results (of different models tried out) in one table (RMSLE, key observations).

Q8. Plot residuals for the best model.

Q9. Explain why the winning model performs better.

Reflection Questions - 1.5 Marks

Q10. Why does RMSLE penalize under-predictions more gently than RMSE?

Q11. What are the trade-offs between model simplicity and predictive power?

Q12. Why can't Linear Regression alone capture time-of-day effects effectively?

Submission Components on Taxila (Only one submission will be allowed. Submission will open on 22 May)

1. Report (Python Notebook): with the answers (theoretical or code) for the above 12 questions.
2. Submission.csv file in required format on test data
3. Proof of executing the Assignment on BITS Lab portal (screenshot in image/ pdf format) In case this is not uploaded, you will lose 0.5 Marks

The submission details will be shared on 22 May EoD.

For any other queries/ clarifications related to the assignment, please write on the Taxila Discussion Forum, after checking existing queries and their answers.

Submission Deadline – **31st May 23:59 PM**

This is a Hard Deadline - No extensions will be provided.

All the Best !!

ML Assignment 1

Submission Instructions

Please read the following instructions carefully before submitting your assignment on the Taxila (e-learn) portal.

Step 1: Prepare the Required Files

1. **Assignment Notebook**

Upload your completed .ipynb notebook containing:

- Your code
- Outputs
- Answers to all assignment questions

2. **submission.csv File** (We will recalculate the rmlse score for your best reported model)

Your prediction file must follow the exact format below:

Required Columns

- datetime
- count_predicted
- Important Requirements
- The file must contain exactly 2 columns: datetime and count_predicted.
- Do not rename the columns.
- Number of rows must exactly match the provided test dataset.
- Do not modify the datetime format or ordering.
- count_predicted must contain only numeric values.
- No missing values or empty cells are allowed.
- Upload only a .csv file (not Excel or any other format).
- The submission.csv file should contain predictions generated using your best performing final model.

3. **Screenshot**

Upload a screenshot showing successful execution of your notebook/code.

Step 2: Verify Your Files Before Submission

- All required files are included
- The notebook runs without errors (outputs should be included in the submission)
- The CSV format is correct
- File names are clear and readable

Final Submission

• Place all required files into a single ZIP folder:

- ✓ .ipynb notebook
- ✓ submission.csv
- ✓ Screenshot

• Upload the ZIP file on the Taxila (e-learn portal) under Assignment 1.

• Late submissions will not be accepted.

• Ensure that the uploaded files are correct before final submission.

All the Best for Your Assignment!

Happy Learning ☺